

AI Acknowledgement, the following AI tools were used in preparing this submission:

Tool	Version	Publisher	URL	Use
Claude	Opus 4.6	Anthropic	https://www.anthropic.com/claude	Main orchestrator: system design, memory architecture, research exploration, literature discovery, draft critique
Claude	Sonnet 4.6	Anthropic	https://www.anthropic.com/claude	Tutoring agent: direct student interaction during pilot sessions
NLTK	3.9	NLTK Project	https://www.nltk.org/	Sentiment analysis, tokenisation, vocabulary analysis of tutoring transcripts
textstat	0.7.4	textstat	https://pypi.org/project/textstat/	Readability scoring (Flesch-Kincaid, Reading Ease) of agent responses
Gemini	3.1	Google DeepMind	https://gemini.google.com/	Research exploration and literature discovery Generate gant-chart
NotebookLM	-	Google	https://notebooklm.google.com/	Interrogating, summarising, and fact-checking research paper PDFs
Tailo	UK	Tailo	https://read.uk.tailoapp.com/	AI-powered research reading tool for literature discovery and paper analysis
OpenClaw	2026.2	OpenClaw	https://github.com/openclaw/openclaw	Multi-agent orchestration platform for pilot study prototype deployment

Multi-Agent AI as Distributed Cognitive Support for ADHD Learners

Background

Attention-deficit/hyperactivity disorder (ADHD) is characterised by differences in the prefrontal cortex, the brain region responsible for executive functions: working memory, impulse control, task-switching, sustained attention and planning (Arnsten, 2009). It impacts approximately 5% of children and adolescents (NICE, 2018). Students with ADHD often experience challenges in the school environment, with working memory deficits leading to comprehensive academic underachievement (Chiu et al., 2021). The students who most need structured revision are therefore the least cognitively equipped to structure it independently, creating a double burden that existing educational technology does not address.

In a 2005 freestyle chess tournament, in which any combination of human and computer could compete, the winners were not the grandmaster or the supercomputer, but two amateur players using three ordinary computers. Kasparov concluded, *"A weak human player plus machine plus a better process is superior to a strong player plus machine plus an inferior process"* (Kasparov, 2017). In other words, the architecture of collaboration matters more than the baseline strength of either component (Dellermann et al., 2019). Even non-agentic adaptive systems, such as DGT Centaur chess computers, have demonstrated positive engagement with ADHD children in chess (Sundaramadhavan et al., 2023); no study has extended this to using multi-agent architecture for structured academic tutoring to support neurodivergent learners.

While this centaur model has transformed fields such as medicine (Topol, 2019) and hybrid intelligence research (Dellermann et al., 2019), it has yet to be meaningfully applied to educational tutoring. The current Edtech-AI developments include pedagogy-driven large language models such as Google's LearnLM (LearnLM Team, 2024), adaptive learning platforms such as Century Tech and Khan Academy, and NLP-based assessment tools such as TeachEdge.ai (TeachEdge, 2026) that automate formative feedback. Concurrently, multi-agent AI frameworks such as Google's Agent Development Kit (Google, 2025) have been developing rapidly though applications tend to be in areas of software engineering and process optimisation tasks. None of these approaches offer a holistic solution to student learning centred on student cognitive needs; they optimise content delivery or assessment but do not scaffold the metacognitive processes (planning, self-monitoring, gap identification) that ADHD learners find most challenging. Does the architecture of AI collaboration matter more than the capability of any single model when supporting ADHD learners? This research proposes a centaur system where an ADHD learner works with an ensemble of AI agents (tutor, examiner, metacognition coach, progress tracker) that shifts cognitive load from the learner to the system.

This research proposal builds directly on work at the University of Leeds, where Alsafari et al. (2024) developed an LLM-powered teaching assistant for the Data Mining and Text Analytics module, evolving from intent-based chatbots to generative AI tutors. The proposal is to extend this trajectory from a single-agent chatbot to a multi-agent architecture, and from a general student population to neurodivergent learners with specific cognitive support needs. In the UK, Le Cunff et al. (2025) at King's College London proposed the FEDIS+R framework identifying six critical areas impacting cognitive load for neurodivergent students in online learning: Format, Environment, Delivery, Instruction, Support, and Research. This proposal implements that theoretical framework through a working multi-agent prototype, translating each of Le Cunff's six areas into specific agent behaviours with measurable outcomes (see pilot study). Internationally, Google's LearnLM represents the current state-of-the-art in pedagogy-driven AI, designed from the ground up to manage cognitive load, inspire active learning, deepen metacognition, stimulate creativity and adapt to the learner, with expert pedagogical raters reporting an average preference strength of 31% over GPT-4o across a diverse set of learning scenarios (LearnLM Team, 2024). However, LearnLM remains a single-model approach; this research asks whether giving the learner a coordinated team of specialised agents, working in synergy, allocated by cognitive need rather than task type, can outperform even the best individual tutor.

Research hypothesis and objectives

Research Hypothesis This research posits that a multi-agent artificial intelligence system, designed according to the centaur principles of human-machine collaboration, will produce measurably better learning outcomes for ADHD learners undertaking GCSE revision than both single-agent tutoring and unaided study, as quantified by supported versus unsupported exam score differentials and improvements in self-correction frequency.

Motivation Three factors motivate this research. Scientifically, a longstanding challenge in educational technology has been Bloom's "Two Sigma" problem (Bloom, 1984): students receiving one-to-one, mastery-based tutoring perform two standard deviations better than those in conventional classrooms. Yet intelligent tutoring systems achieve only $d=0.76$ (VanLehn, 2011), far below Bloom's ceiling, partly because managing cognitive load (Sweller, 1988) alongside content delivery exceeds what a single system can optimally handle. For ADHD learners, whose executive functioning is easily overwhelmed, a single monolithic AI tutor cannot simultaneously provide subject-matter expertise, monitor cognitive load, and offer metacognitive scaffolding. Replicating optimal human tutoring requires a coordinated ecosystem of specialists.

Professional context: The researcher's work in education provides regular engagement with students facing these systemic barriers.

Timeliness and Novelty 2024 represented a watershed moment for multi-agent AI, with the maturation of frameworks such as AutoGen (Wu et al., 2024) and MetaGPT (Hong et al., 2024). Concurrently, NHS ADHD assessment waiting times exceed seven years in some regions (ADHD UK, 2023). The convergence of capable AI orchestration tools with this clinical and educational need creates a unique window. The novelty lies in applying the centaur model to education: combining multi-agent orchestration, ADHD-specific cognitive scaffolding, and text analytics into a unified architecture. No existing work combines multi-agent AI orchestration with ADHD-specific cognitive load management in educational tutoring; this proposal addresses that gap.

Overall Aim

To design, implement, and evaluate a centaur-model multi-agent AI tutoring system that provides ADHD-specific cognitive scaffolding across multiple GCSE subjects, reconceptualising multi-agent AI as a distributed external prefrontal cortex where each agent replaces a specific executive function that ADHD impairs.

Measurable Objectives

1. Design a multi-agent architecture with four specialised AI roles (subject tutor, examiner, metacognition coach, progress tracker), mapped to executive function deficit models (Barkley, 2012) and calibrated to manage cognitive load.
2. Implement NLP-based transcript analysis using NLTK to detect comprehension gaps, classify error patterns (procedural, conceptual, careless), and monitor self-correction frequency.
3. Conduct a full study with a small group of 4-6 ADHD learners revising across GCSE subjects, comparing supported (multi-agent) versus unsupported performance on past exam questions.
4. Measure the supported versus unsupported grade gap, targeting a reduction from an estimated baseline of two grades to one grade or less.
5. Produce a replicable CRISP-DM evaluation framework benchmarking multi-agent output against single-model baselines across token cost, pedagogical accuracy, and learner engagement.

Importance and contribution to knowledge

Research Gap: Despite four decades of intelligent tutoring systems research and the recent emergence of multi-agent AI architecture, no existing work addresses the intersection of these fields for neurodivergent learners. The ITS literature, from early rule-based systems to contemporary LLM-powered tutors, has overwhelmingly assumed a neurotypical learner profile. Adaptive platforms such as ALEKS, DreamBox, and Century Tech adjust content difficulty based on performance metrics, but none account for the executive function deficits that fundamentally alter how ADHD learners process, retain, and engage with information (Barkley, 2012). Simultaneously, multi-agent orchestration frameworks such as AutoGen (Wu et al., 2024) and MetaGPT (Hong et al., 2024) have demonstrated powerful autonomous collaboration in software engineering and scientific research, but have not yet been directed towards educational accessibility. This proposal bridges that gap, contributing a novel architectural framework in which multi-agent design is driven not solely by task complexity, but by the precise cognitive profile of the human collaborator.

Beneficiaries and Socioeconomic Impact:

Most directly, ADHD learners, representing approximately 5% of school-age children globally (NICE, 2018), stand to gain access to a tutoring system that actively compensates for their specific cognitive challenges. With NHS ADHD assessment waiting times now exceeding seven years in some regions (ADHD UK, 2023), many learners remain undiagnosed and unsupported; a system designed around universal executive function principles, rather than requiring a formal diagnosis, could support a far wider population.

Teachers and SENDCos will benefit from individualised scaffolding without additional staffing costs. EdTech developers will gain a replicable design framework for neurodivergent-aware AI applications. Findings could also inform policymakers, contributing to DfE guidance on AI in SEND provision and supporting the SEND review's recommendations for technology-enabled inclusive education. In the UK, students with ADHD are more likely to face exclusions, achieve below predicted grades, and require costly additional support (Department for Education, 2024). A scalable AI-driven system could reduce these costs while improving outcomes.

Contributions to Knowledge:

This research makes four distinct contributions. First, it provides the first empirical application of Kasparov's centaur model to structured academic tutoring; adaptive AI systems (such as the DGT Centaur chess computers) have demonstrated positive engagement and a reduction in behavioural issues among children with ADHD (Sundaramadhavan et al., 2023), though such human-AI collaborative models have not yet been applied to structured academic tutoring with measurable outcomes. Second, it contributes a replicable methodology for matching AI agent roles to specific cognitive deficits, providing researchers across educational psychology, computer science, and special education with a shared framework transferable to other neurodivergent profiles. Third, by combining NLP-driven transcript analysis with multi-agent orchestration, it demonstrates how data mining and text analytics can be applied to educational cognitive support, connecting three fields into a unified programme. Fourth, it synthesises Bloom's (1984) two-sigma problem, VanLehn's (2011) $d=0.76$ finding, and Sweller's (1988) Cognitive Load Theory for neurodivergent populations for the first time, testing whether a multi-agent architecture can meet or exceed these benchmarks for ADHD learners.

If validated, this approach could fundamentally shift how educational technology accommodates cognitive diversity, moving from one-size-fits-all platforms to systems that adapt their internal structure to the learner's neurological profile.

Critical Analysis of Existing Solutions

Approach 1: Traditional Intelligent Tutoring Systems

Purpose and fit: Intelligent tutoring systems (ITS) such as Century Tech, DrFrostMaths, ALEKS, and DreamBox represent the most mature category of adaptive learning technology, dynamically adjusting content difficulty to student progress. VanLehn's (2011) comprehensive meta-analysis established that ITS achieves an effect size of $d=0.76$ over conventional instruction, nearly matching human tutoring at $d=0.79$. This narrow gap made ITS appear a cost-effective alternative to one-to-one tutoring and a natural starting point for this project.

Technical limitations: A 2025 systematic review of AI-driven ITS in K-12 education confirmed that effects remain generally positive, though mitigated and often not statistically significant when compared to non-intelligent tutoring systems (Chrysafiadi et al., 2025). More critically, the entire evidence base assumes a neurotypical learner population. None of the systems reviewed in either VanLehn's meta-analysis or the 2025 systematic review incorporate adaptations for executive function deficits. Le Cunff et al.'s (2024) systematic review, the first to examine the relationship between neurodiversity and cognitive load in online learning, found that ADHD learners experience and process cognitive load through fundamentally different neurophysiological pathways than their neurotypical peers. A complementary framework paper further confirmed that standard Cognitive Load Theory research has not been validated for neurodivergent populations (Le Cunff et al., 2025).

Rejection: The architecture of traditional ITS, a single adaptive model adjusting content difficulty based on performance metrics alone, cannot account for the working memory limitations, impulse control difficulties, and task-switching challenges that define the ADHD cognitive profile (Barkley, 2012). ITS is therefore rejected: its proven effectiveness for neurotypical populations cannot be assumed to transfer to the target population of this research.

Approach 2: Single-LLM Tutoring

Purpose and fit: Large language model tutors, most notably ChatGPT and Khan Academy's Khanmigo (over 2 million users worldwide by 2024-25) introduced a qualitative shift in AI tutoring capability (Khan Academy, 2025). Deng et al.'s (2024) systematic review and meta-analysis found that ChatGPT enhances academic performance and reduces perceived mental effort, while a controlled study with 274 participants demonstrated that ChatGPT-generated help produces learning gains equivalent to human tutor-authored help on mathematics problem-solving (Lin et al., 2024). Google's LearnLM represents the current state of the art, achieving preference ratings of +31% over GPT-4o, +11% over Claude 3.5, and +13% over its Gemini 1.5 Pro base model as rated by expert pedagogy evaluators (LearnLM Team, 2024). These results make single-LLM tutoring a credible candidate for this project.

Technical limitations: Even LearnLM, the strongest single-model tutor available, suffers from three critical limitations when applied to ADHD learners. First, a single model cannot simultaneously fulfil the distinct pedagogical roles required for ADHD support: deep subject-matter expertise, cognitive load monitoring, metacognitive scaffolding, and longitudinal progress tracking. These functions compete for prompt context and produce inconsistent prioritisation within a monolithic architecture. Second, single-LLM sessions are inherently stateless: ChatGPT cannot maintain a persistent learner model across sessions, making it fundamentally incapable of implementing spaced retrieval schedules or identifying longitudinal error patterns. For ADHD learners, who struggle with precisely these planning and self-monitoring tasks, every session starts from zero. Third, Stadler et al. (2024) demonstrated that LLMs reduce germane cognitive load through reduced need to make inferences, resulting in reduced processing depth and worse memorisation. This is particularly damaging for ADHD learners, who already struggle with deep processing and effortful encoding.

Rejection: Single-LLM tutoring is rejected: its monolithic architecture cannot separate the cognitive support functions this project requires, its statelessness prevents the longitudinal tracking essential for spaced retrieval and gap identification, and its tendency to reduce effortful processing directly conflicts with ADHD learning needs.

Approach 3: Multi-Agent AI Frameworks

Purpose and fit: Multi-agent orchestration frameworks represent the most architecturally promising approach. AutoGen (Wu et al., 2024), presented at ICLR 2024, enables customisable multi-agent conversations with specialised roles, while MetaGPT (Hong et al., 2024) demonstrated sophisticated collaboration through structured communication protocols. The educational application closest to this project, EvaAI (Lagakis and Demetriadis, 2024), applied AutoGen to automated MOOC grading, mimicking human scoring processes through multi-agent collaboration.

Technical limitations: Multi-agent systems remain fragile at scale: Cemri et al. (2025) analysed 1,600+ execution traces across seven popular frameworks and found failure rates reaching 86.7%, identifying 14 unique failure modes including inter-agent communication breakdowns, memory management failures, and premature task termination. Beyond reliability, every existing multi-agent implementation allocates agents *by task type rather than by cognitive need*. MetaGPT assigns roles as product manager, engineer, and architect, mirroring organisational hierarchies, not learner profiles (Hong et al., 2024). Even EvaAI, the closest educational application, addresses assessment rather than tutoring and makes no consideration for neurodiversity (Lagakis and Demetriadis, 2024).

Rejection: Multi-agent baselines are rejected as insufficient: the orchestration capability is proven, but the architecture suffers from systemic reliability issues, and the allocation principle must be fundamentally redesigned around cognitive need rather than task type.

Contradictory Findings and Resolution

A significant contradiction emerges between the evidence bases of the first two approaches. VanLehn (2011) established that AI tutoring systems achieve near-human effectiveness ($d=0.76$), implying that technology can successfully replicate the benefits of expert tutoring. Yet Stadler et al. (2024) found that LLM-based tutoring actively compromises learning depth by reducing the effortful processing required for durable memory formation. If AI tutoring works, why does it simultaneously undermine the cognitive processes that make learning stick?

This contradiction is resolved by this proposal's central argument: it is not the presence of AI that determines effectiveness, but the architecture of its deployment. A monolithic system that uniformly reduces cognitive effort produces the shallow processing Stadler et al. documented. A multi-agent system that deliberately manages which type of support is delivered, by which specialised agent, and at which moment in the learning process, can achieve VanLehn's demonstrated effectiveness while preserving the effortful engagement that Stadler et al. show is essential. The metacognition coach agent exists precisely to maintain productive struggle, preventing the "cognitive ease at a cost" trap by ensuring that scaffolding supports the learner without bypassing the cognitive work required for deep and meaningful learning.

Pilot study

Case Study and Data Acquisition

A single participant, Student A (Year 11, suspected ADHD-Combined), completed seven tutoring sessions totalling 4.8 hours across five subjects (Mathematics, Chemistry, History, Design Technology, English) over five days (26 February to 2 March 2026). Sessions were delivered through a live prototype deployed on the OpenClaw orchestration platform, generating 182 timestamped messages (51 student, 131 assistant) for computational analysis.

Prototype Design and Rationale

The prototype extended the task-driven ReAct baseline model with a fundamental redesign: agent roles were allocated by cognitive need rather than task type. Four specialised agents operated under a central orchestrator:

- (1) a Subject Tutor delivering curriculum-aligned content mapped to 4,537 past paper questions across 645 specification points
- (2) an Examiner administering retrieval practice using real mark schemes to counteract ADHD self-monitoring deficits
- (3) a Metacognition Coach providing scaffolding for task initiation and self-regulation
- (4) a Progress Tracker maintaining a persistent cross-session learner model that offloaded the meta-tasks of revision management (planning, gap identification, spaced retrieval scheduling) from the student to the system.

A key design decision concerned memory architecture. Early sessions revealed a "restart load": Student A had to re-explain prior coverage each session, compounding ADHD task-initiation difficulties. This led to a four-layer memory structure: session-level context (conversational coherence), daily logs (session outcomes), curated long-term memory (persistent learner profiles), and a structured database (645 specification points, 79 topic confidence entries, 46 knowledge gaps). By injecting prior session context automatically, the system eliminated re-initiation friction, an architectural decision that emerged from observing the student rather than from theory. The database enabled a closed-loop adaptive cycle: question-level results updated confidence scores, which drove automated spaced repetition scheduling, effectively externalising the meta-cognitive planning that ADHD most impairs.

These agents produced synergistic behaviours that no single agent could replicate. When Student A requested "explain it more simpler" during Chemistry, the orchestrator routed through the Metacognition Coach first, which assessed the difficulty as linguistic rather than conceptual, before the Subject Tutor adapted its register, reducing a technical explanation to: "It's a straw that measures. You suck up liquid, it holds exactly 25cm³, you squeeze it into the flask. Done." A monolithic system would process this as one undifferentiated response; the multi-agent architecture separates diagnosis from adaptation. The system maintained persistent memory across all seven sessions, enabling cumulative learner modelling.

Evaluation Results

Transcripts were analysed using NLTK (VADER sentiment), textstat (readability), and LLM-based classifiers (error type, pedagogical features), directly applying text analytics methods covered in the Data Mining and Text Analytics module:

Metric	Result	Significance
Mean Flesch-Kincaid grade	6.1 (range: 3.7 Maths to 7.2 Design Technology)	Language pitched below GCSE level to reduce cognitive load
Errors identified	Top 3: 28.6% partial knowledge, 25.0% procedural, 7.1% conceptual	Majority issues are partial or procedural and addressable through targeted retrieval practice
Self-correction rate	28.6%	Emerging metacognitive awareness without explicit prompting

Metric	Result	Significance
VADER sentiment trajectory	-0.084 (Q1) to +0.027 (Q4)	Shift from uncertainty to neutral-positive engagement
Frustration markers	2.0% of messages (despite 28 errors)	Psychological safety maintained throughout
Formative assessment	37.4% of responses	Personalisation 32.1%, Socratic questioning 25.2%

Unexpected Findings 1: Student linguistic capability exceeded expectations. Student A's vocabulary diversity was notably high (type-token ratio 0.727 across 170 unique words), contradicting the assumption that ADHD learners produce linguistically poor responses. This finding supports Barkley's (2012) distinction between performance deficits and knowledge gaps: the executive function deficit manifests in sustained attention and task management, not in linguistic capability. The learner's difficulty was never "not knowing the words" but rather sustaining the structured effort to deploy that knowledge effectively. This has direct implications for system design: the architecture should target organisational scaffolding rather than linguistic simplification.

Unexpected Findings 2: Low lexical diversity in the assistant was a strength, not a weakness. The assistant's type-token ratio (0.316) was deliberately low, with repeated structural phrases ("Write this down", "Send the next page") providing predictability across sessions. Rather than indicating poor output quality, as standard NLP evaluation would suggest, this repetition appeared to benefit engagement by providing the routine and structure that compensate for ADHD executive dysfunction. This contradicts the general assumption that higher lexical diversity indicates higher quality output and suggests that NLP evaluation metrics for educational AI must be calibrated differently for neurodivergent learners where consistency can imply better quality.

Limitations

This pilot involved a single participant over five days, limiting generalisability. The prototype implemented the four cognitive roles within a single orchestrated session using structured role-switching prompts, rather than as independently deployed agent instances; while all described behaviours were produced and evidenced in transcripts, the full study will deploy discrete agents to enable controlled single-agent versus multi-agent comparison. VADER likely underrepresents positive engagement in factual educational dialogue. The single-model comparison was hypothetical rather than empirical; the full study will conduct controlled A/B testing using Google's LearnLM (LearnLM Team, 2024), the current state-of-the-art pedagogy-aware single-model tutor, as the baseline comparator. Following the pilot, the system was extended into an eight-week GCSE exam revision programme covering all eight examination subjects, with automated scheduling and spaced repetition across 645 specification points, demonstrating the architecture's readiness for full deployment.

Programme and methodology

This research adopts CRISP-DM for its iterative structure, which forces alignment between technical development and educational need. CRISP-DM is appropriate for this project because its iterative cycle mirrors the adaptive nature of ADHD-responsive tutoring: learner needs are not fully knowable in advance and the system must evolve based on observed behaviour. Unlike waterfall methodologies, CRISP-DM's feedback loop between evaluation and data understanding enables the system to refine its agent configurations based on transcript analysis from each tutoring cycle. Two module methods are integrated: NLTK for transcript analysis and LLM-based classification for error categorisation and pedagogical feature detection. Progress is monitored through fortnightly supervisor meetings, a shared Gantt chart updated weekly, and automated session logging via gauss.db which provides real-time visibility into participant engagement and data collection completeness.

Phase 1: Business Understanding – Educational Needs Analysis (Weeks 1-2)

Establish quantitative baselines for a small group of students (target n=4-6) with ADHD or suspected ADHD across GCSE subjects using past paper scores. Document each learner's cognitive profile, curriculum constraints (Edexcel, AQA, OCR), and success criteria (grade gap reduction from two grades to one or less). Participants are recruited from a single school with SENDCo support.

Risks: (1) Baseline performance may be inconsistent due to ADHD variability in attention, producing unreliable benchmarks. (2) Small sample size limits statistical power and generalisability. (3) Stakeholder requirements may shift as learners' needs evolve, requiring scope renegotiation.

Phase 2: Data Understanding (Weeks 3-4)

Tutoring transcripts form the primary dataset: timestamped, speaker-labelled text exchanges per participant. The pilot produced 182 messages across 4.8 hours for one student; the full study projects 1,500-2,000 messages per participant. Data quality assessment includes message type identification, annotation conventions for error classification, and cataloguing past paper questions mapped to specification points.

Risks: (1) Image-based responses require OCR preprocessing, risking 15-20% data loss. (2) System metadata may contaminate text analysis if not rigorously cleaned. (3) Uneven subject distribution may produce insufficient transcript volume in avoided subjects.

Phase 3: Data Preparation (Weeks 5-6)

Transcripts are cleaned, tokenised, and feature-extracted using NLTK and textstat. Features include: Flesch-Kincaid readability scores, VADER sentiment scores, error type classifications, and pedagogical feature labels (formative assessment, scaffolding, Socratic questioning). Anonymisation replaces identifying information with pseudonyms.

Risks: (1) VADER, optimised for social media, may misclassify neutral educational responses as lacking engagement. (2) Error type classification requires human judgment at procedural-conceptual boundaries, introducing subjectivity. (3) Anonymisation must satisfy ethical requirements while preserving contextual detail for analysis.

Phase 4: Modelling (Weeks 7-12)

The multi-agent system is deployed on OpenClaw using a coordinator-worker pattern. Four agents map to ADHD executive function deficits (Barkley, 2012): Subject Tutor (content delivery), Examiner (retrieval practice), Metacognition Coach (task initiation scaffolding), and Progress Tracker (persistent learner model). The full study adds a fifth agent, a **Wellbeing Monitor**, to track emotional state and escalate to safeguarding pathways when needed. Two module methods are integrated: NLTK for transcript analysis and LLM-based classifiers for error categorisation and pedagogical feature detection.

Risks: (1) Agent coordination latency may disrupt real-time tutoring if orchestration overhead exceeds acceptable response times. (2) Prompt engineering is inherently brittle; small changes can disproportionately shift output quality. (3) The persistent memory system may accumulate incorrect learner model assumptions, requiring periodic validation.

Phase 5: Evaluation (Weeks 13-20)

Quantitative measures: supported vs unsupported grade differential, error type distribution shift, engagement metrics, and token cost comparison. Qualitative evaluation uses LLM-as-Judge ([Zheng et al., 2023](#)) to assess pedagogical quality against rubrics for scaffolding, cognitive load management, and personalisation. The A/B comparison becomes empirical: matched questions under single-model (LearnLM) and multi-agent conditions, counterbalanced for learning effects.

Risks: (1) Small sample size limits inferential statistical analysis. (2) LLM-as-Judge may exhibit biases favouring verbose responses. (3) Improvement may be confounded by maturation, practice, or external tutoring.

Phase 6: Deployment (Weeks 21-24)

Produce a replicable deployment framework: agent architecture specification, prompt templates, evaluation rubrics, cost-benefit analysis, and a user guide for teachers and SENDCos. Findings synthesised for conference submission.

Risks: (1) School IT infrastructure may not support API connectivity for live orchestration. (2) Teacher adoption requires CPD investment exceeding available time. (3) GDPR and DfE guidance may constrain transcript storage, requiring architectural modifications.

Milestones and Deliverables

Week	Milestone	Deliverable
2	Baselines established	Learner profiles, baseline scores
4	Data pipeline validated	Cleaned transcript corpus, annotation schema
6	Feature extraction complete	NLTK/textstat feature set per participant
12	Full deployment operational	Multi-agent system, 4-6 participants active
16	A/B comparison complete	Supported vs unsupported score dataset
20	Evaluation report drafted	Statistical analysis, LLM-as-Judge results
24	Final submission	Thesis, open-source toolkit, conference paper

Team Roles and Stakeholders

Role	Responsibility
Researcher (author)	System design, data analysis, evaluation, report writing
Student participants (n=4-6)	Tutoring sessions, feedback on system usability
Module supervisor	Methodological guidance, marking criteria alignment
School SENDCo (advisory)	ADHD learner profile validation, deployment feasibility
External examiner	Independent quality assurance of evaluation framework

Workplan diagram

Research Work Programme: CRISP-DM Methodology

Centaur Systems for Cognitive Support in GCSE Revision for ADHD Learners

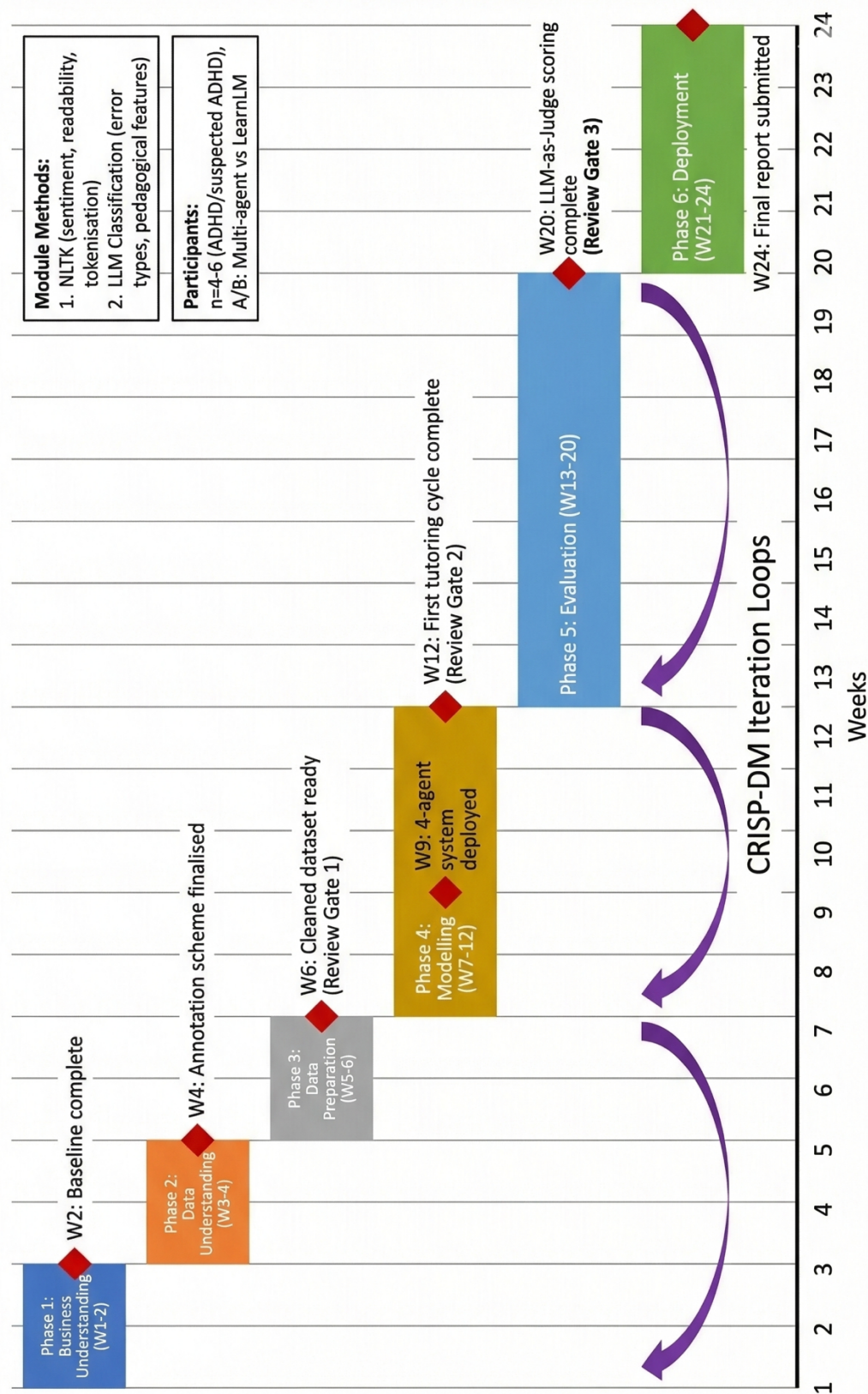


Figure 1: Research work programme following CRISP-DM methodology. Purple vertical lines indicate review gates (W6, W12, W20). Purple arrows indicate iteration loops where findings feed back into earlier phases. Red diamonds mark key milestones and delivery

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Appendix: Tools, Prompts, and Error Analysis

A1. Tools Used in the Pilot Study

NLTK (Natural Language Toolkit) and textstat — Transcript Analysis NLTK was used to perform computational analysis on the 182-message tutoring transcript corpus: VADER sentiment analysis to track learner emotional trajectory, tokenisation for vocabulary analysis (type-token ratio), and keyword-based frustration/engagement detection. The textstat library computed Flesch-Kincaid grade levels and Reading Ease scores for assistant responses exceeding 15 words (n=120 of 131); shorter responses were excluded as textstat produces unreliable scores on very short text.

Example:

```
from nltk.sentiment.vader import SentimentIntensityAnalyzer
sia = SentimentIntensityAnalyzer()
user_sentiments = []
for m in real_user_msgs:
    scores = sia.polarity_scores(m['clean_text'])
    user_sentiments.append({'compound': scores['compound'], 'pos':
scores['pos'],
                           'neg': scores['neg'], 'neu': scores['neu']})

# Readability: 5 metrics on responses >15 words
asst_texts = [m['clean_text'] for m in real_asst_msgs if
len(m['clean_text'].split()) > 15]
fk_scores = [textstat.flesch_kincaid_grade(t) for t in asst_texts]
fre_scores = [textstat.flesch_reading_ease(t) for t in asst_texts]
fog_scores = [textstat.gunning_fog(t) for t in asst_texts]
smog_scores = [textstat.smog_index(t) for t in asst_texts if
len(sent_tokenize(t)) >= 3]
ari_scores = [textstat.automated_readability_index(t) for t in asst_texts]

# Keyword-based frustration/engagement detection
frustration_keywords = ['confused', "don't get", "don't understand", 'stuck',
                        'hard', 'difficult', "can't", 'lost', 'frustrated']
frust = sum(1 for m in real_user_msgs
            if any(fw in m['clean_text'].lower() for fw in
frustration_keywords))
```

Results: Sentiment trajectory across session quarters:

Quarter	Compound Score	n	Interpretation
Q1	-0.084	12	Initial hesitancy
Q2	-0.004	12	Settling in
Q3	+0.092	12	Growing confidence
Q4	+0.027	12	Neutral-positive engagement

A residual fifth quarter (n=3) was excluded due to insufficient sample size. Frustration markers appeared in only 2.0% of messages despite 28 error instances, with engagement markers in 5.9%, indicating the multi-agent system maintained psychological safety throughout the pilot.

Readability by subject:

Subject	FK Grade	Reading Ease	n
Mathematics	3.7	88.4	26
History	6.5	70.3	13
Chemistry	6.7	70.9	38
English	6.8	71.1	4
Design Technology	7.2	65.3	33
Overall	6.1	73.4	120

The system pitched language below formal GCSE level across all subjects, making content cognitively accessible for an ADHD learner. Mathematics scored lowest (FK 3.7) due to numerical working; Design Technology scored highest (FK 7.2) due to multi-syllable technical terminology.

OpenClaw Multi-Agent Platform — Prototype Deployment

The centaur tutoring system was deployed on OpenClaw, an orchestration platform supporting coordinator-worker agent patterns with persistent cross-session memory. Four cognitive roles (Subject Tutor, Examiner, Metacognition Coach, Progress Tracker) operated under a central orchestrator.

Example interaction (History, Cold War session, 26 February 2026): Student A answered a source analysis question: "This shows Americans have the power to gain more soldiers against Cuba, and the cartoon is like an advertisement." The system validated partial knowledge before correcting, then provided the mark scheme formula: "Good start — you've spotted that America is using money/power. But we need to push further for marks. The mark scheme says you need: 1) What the source shows (content), 2) Your own knowledge to explain WHY it's critical. [...] Write this down: Q1 formula: Source shows [thing] + I know [context/event] = explains why it's critical." Student A's subsequent attempt applied the formula correctly, demonstrating scaffolded learning transfer within a single exchange.

Keyword-Based Classification — Pedagogical Feature Detection

Pedagogical features in assistant responses were detected using keyword pattern matching against seven categories, each defined by curated indicator phrases drawn from the transcripts. This approach was chosen for reproducibility and transparency: every detection is traceable to a specific keyword match.

Example:

```
formative_p = ['mark scheme', 'marks', '/1', '/2', '/3', '/4', 'running score']
scaffolding_p = ['let me', 'first', 'step 1', 'step 2', 'break it down']
socratic_p = ['what do you think', 'have a go', 'your turn', 'cold recall']

for feature, patterns in features.items():
    count = sum(1 for m in real_asst_msgs
                 if any(p in m['clean_text'].lower() for p in patterns))
```

Result: Formative assessment 37.4%, personalisation 32.1%, Socratic questioning 25.2%, motivational reinforcement 24.4%, cognitive load management 22.1%, scaffolding 20.6%, adaptive difficulty 3.1%. The low adaptive difficulty rate (4 instances) suggests the system's default language

level (FK 6.1) was appropriately calibrated, requiring explicit simplification only when partial answers indicated a comprehension gap.

A2. Tools Used for Background Research

Google Scholar, Semantic Scholar, Tailo, NotebookLM, Gemini 3.1, and Claude Opus 4.6

Systematic searches used queries including "ADHD intelligent tutoring systems," "multi-agent educational AI," "cognitive load neurodevelopmental," and "LLM tutoring effectiveness 2024." Google Scholar provided broad coverage; Semantic Scholar enabled filtering by citation count and recency. Tailo and NotebookLM were used to interrogate, summarise, and fact-check research paper PDFs. Claude Opus 4.6 and Gemini 3.1 explored research directions and identified contradictions. Key papers discovered: Le Cunff et al. (2024), Stadler et al. (2024).

Example prompt (Tailo):

Can a multi-agent AI system designed around the centaur model of human-machine collaboration with agents allocated by cognitive needs produce better learning outcomes for ADHD learners?

Result (AI-generated key findings):

- "Multi-agent AI systems, particularly those employing a centaur model and allocating agents based on cognitive needs, show promise in enhancing learning outcomes for ADHD learners." (Sundaramadhavan et al., 2023)
- "Adaptive chess computers have demonstrated positive engagement with children with ADHD, reducing behavioral issues." (Sundaramadhavan et al., 2023)
- "Digital game technologies can help mask disruptive behaviors and maintain cognitive engagement in classroom settings." (Sundaramadhavan et al., 2023)

This confirmed Sundaramadhavan et al. (2023) as the closest precedent, limited to gaming contexts rather than structured academic tutoring.

Example prompt (Claude Opus 4.6):

VanLehn (2011) found ITS effect size of 0.76 standard deviations. Are there 2024 papers that contradict or complicate this for neurodivergent populations?

Result: Identified the Stadler et al. (2024) "cognitive ease at a cost" paper and Le Cunff et al. (2024) scoping review on cognitive load in neurodevelopmental conditions, both central to the critical analysis argument.

A3. Tools Used to Draft and Improve the Report

Claude (Opus 4.6, Anthropic) — Draft Critique and Improvement

Claude Opus 4.6 was used to critique the first draft, identify structural weaknesses, and fact-check claims against cited sources. It was not used to generate written content.

Prompt:

Please critique the following research proposal draft for an MSc AI module.

Evaluate: (1) structural coherence, (2) evidence quality, (3) tone and style, (4) three biggest vulnerabilities. [Full draft submitted]

Critique (summarised):

1. **Structural strength confirmed:** The funnel structure (gap, three approaches, contradiction, resolution, pilot, methodology) was identified as coherent and well-paced.
2. **Evidence gap identified:** The single-model comparison in the pilot study was flagged as hypothetical rather than empirical. Recommendation: acknowledge as limitation with controlled A/B testing in full study.
3. **Tone appropriate:** Academic register maintained, technical terminology precise and defined.
4. **Vulnerabilities:** (a) Single-participant design limits generalisability. (b) VADER may underrepresent engagement in educational dialogue. (c) Centaur education claim should be hedged as emerging preprint work may exist.

Action: (a) and (b) already documented in Pilot Study limitations. (c) led to a preprint search confirming no published application of the centaur model to ADHD-specific educational support as of February 2026.

A4. Error Documentation

Error 1: VADER Sentiment Misclassification of Educational Dialogue *Description:* VADER classified 70.6% of student messages as neutral (compound score near zero), despite many containing correct, engaged answers. *Cause:* VADER was trained on social media text (tweets, product reviews) where emotional valence is explicit. Educational dialogue, where a student answering "the atomic number is the number of protons" is genuinely engaged but linguistically neutral, falls outside VADER's training distribution. *Resolution:* Supplemented with keyword-based engagement detection (engagement markers 5.9%, frustration markers 2.0%). Sentiment trajectory retained as a relative measure. Flagged for replacement with a fine-tuned educational sentiment classifier in the full study.

Error 2: Transcript Contamination from System Metadata *Description:* Raw transcripts contained embedded Telegram formatting codes, media file paths, agent routing labels, and session boundary markers, inflating word counts and corrupting readability scores. *Cause:* The extraction pipeline exported full message objects rather than cleaned text content. System-generated fields were not separated from human-generated text during initial export. *Resolution:* A custom regex cleaning script stripped all non-content fields. Word counts dropped from an inflated 12,700 to the accurate 8,616 after cleaning. All subsequent analysis used the cleaned corpus. The cleaning script is documented in the pilot study scripts directory for reproducibility.

Error 3: Role Bleed in Single-Session Architecture *Description:* The four cognitive roles operated within a single orchestrated session using role-switching prompts rather than discrete agent instances. When the student requested metacognitive support (e.g., "Explain it more simpler"), the system sometimes delivered content simplification rather than first diagnosing whether the difficulty was linguistic, conceptual, or motivational. *Cause:* No explicit routing logic between roles; the model defaulted to content delivery for ambiguous messages. For example, message [146] received a content simplification ("It's a straw that measures") rather than a metacognitive prompt asking the student to identify what was unclear. *Resolution:* Informed the full study design: discrete agent instances with two-stage intent classification (content vs. strategy vs. emotional need) before routing. For ambiguous messages, the Metacognition Coach receives first, then escalates content questions to the Subject Tutor.